



JUSTIN RANKIN

SOFTWARE ENGINEER · TYPESCRIPT · REACT · NODE.JS

Front-end architecture, TypeScript systems, and tooling-led engineering for products that need to scale cleanly.

CONTACT

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PROFILE

Software engineer focused on front-end architecture, TypeScript-first systems, and developer experience. Over fifteen years building for the web, from early full-stack freelance work through agency, startup, and large enterprise environments across Canada, Spain, and Australia.

Spent the last five years at Sage, working on enterprise SaaS products within a large-scale micro-frontend architecture. Collaborated across international teams on shared UI systems, reusable component libraries, internal platform improvements, and front-end modernization. Increasing focus on maintainability, DX tooling, and integrating AI-assisted development workflows into modern TypeScript engineering.

Outside of work, actively building and maintaining a personal open-source TypeScript tooling ecosystem: shared linting and formatting configurations, CLI developer tools, a project generator and codemod toolkit, dependency governance, and a design system built on PandaCSS and Ark UI. All published via GitHub Packages.

Preference for explicit architecture over implicit convention, strong typing, self-documenting code, long-term maintainability, and systems that are readable and scalable by default.

TECHNOLOGIES

Core

TypeScript JavaScript React Node.js

Architecture & Patterns

Micro-frontends Monorepo (pnpm, Turborepo)
Shared component systems Design systems ESM-first modules
@tanstack/query

WORK EXPERIENCE

Sage

Software Engineer

2021 - 2026 · Barcelona, Spain

Front-end and full-stack development for enterprise SaaS products within Sage Small Business Segment (SBS), focused on accounting, finance, invoicing, and administrative platforms, dashboards, and shared front-end infrastructure.

Worked within a large-scale micro-frontend ecosystem spanning international squads across Europe, South Africa, Canada, and the United States. Collaborated across front-end, back-end, architecture, UX, product, and shared platform teams on customer-facing products, internal systems, and production prototypes.

Developed reusable front-end systems, shared UI functionality, and scalable component patterns using TypeScript, React, Node.js, internal Sage Design System tooling, and the Carbon component ecosystem. Front-end work was tightly integrated with backend-developed RESTful API services, consuming defined API contracts for data fetching and mutation, with @tanstack/query used for client-side caching and UI performance optimization.

Contributed to modernization and long-term maintainability initiatives across front-end platforms, including migration and refactor work involving shared dependencies, front-end form architecture, Node.js upgrades, security patching, accessibility improvements, and reduction of legacy technical debt. Unit, end-to-end, and integration testing as standard practice, with strict code coverage requirements across front-end applications.

Front-end applications were containerized with Docker and deployed to AWS-hosted infrastructure; container-level authentication credentials against AWS services were configured and maintained as part of front-end delivery work, with console-level infrastructure and service configuration owned by a dedicated shared-platform squad.

Strong focus on engineering consistency, explicit architecture, developer experience, and maintainable patterns across large enterprise codebases. Daily involvement in pull-request reviews, refactoring initiatives, and front-end standards discussions across squads and shared engineering channels.

Exploration and integration of AI-assisted and agentic development workflows, including complex debugging, task analysis, test generation, and automation. Independent research into modern AI tooling applied selectively within professional engineering context.

TypeScript React Node.js Micro-frontends Design systems
Shared component systems RESTful API integration @tanstack/query
Docker AWS-hosted containerised deployment CI/CD workflows
GitHub workflows Accessibility standards Enterprise SaaS architecture

Front-end & UI

Accessibility (WCAG, ARIA)

Headless UI (Ark UI, Radix UI)

React Hook Form

Schema validation (Zod)

CSS-in-JS (PandaCSS, Emotion)

SASS/CSS

Responsive & mobile-first

i18n / Localization

Tooling & DX

OXC (oxlint, oxfmt)

ESLint

Vite

tsdown

Vitest

@testing-library

CI/CD (TeamCity, GitHub Actions)

Git workflows

Docker

GitHub Packages

pnpm

OXC, tsdown, and Vite/Rolldown are part of the emerging Rust-native JavaScript toolchain.

CLI & Developer Tools

Custom CLI (Commander, @clack/prompts)

TUI rendering

Codemod tooling

Dependency governance

AI & Agentic Development

AI-assisted engineering workflows

Agentic coding (Claude Code, Cursor, Copilot)

LLM integration (Anthropic, Ollama)

MCP server development

Structured knowledge pipelines

Prompt & system instruction design

Testing

Vitest

@testing-library/react

Jest

Playwright

Server & APIs

Hono

RESTful APIs

Drizzle ORM

SQLite

MySQL

MongoDB

Platforms & Infrastructure

Linux

macOS

AWS

GitHub Actions

TeamCity CI/CD

ENGINEERING PHILOSOPHY

Explicit architecture, self-documenting code: Clear naming, semantic conventions, visible structure, readable intent. JSDoc for complex logic and AST usage.

Strong typing: TypeScript for safety, documentation, and developer experience in one.

Maintainability: Reusable code with limited, purposeful abstraction. Easy to read, easy to navigate, easy to change next year.

Developer experience: Tooling, linting, CI, and onboarding friction directly affect what ships.

ESM-first, monorepo-native: Modern module resolution and workspace architecture as defaults.

eDreams ODIGEO

Front-end Developer

2019 - 2020 · Barcelona, Spain

Front-end development for one of Europe’s largest online travel platforms, working on hotel and flight booking flows across desktop and mobile experiences. Collaboration within agile product pods on feature delivery, integrations, testing workflows, refactoring, and continuous integration across a shared front-end environment.

Cross-team engineering workflows involving code reviews, QA processes, release pipelines, and front-end application architecture across an international product organization.

JavaScript

Node.js

RESTful APIs

Docker

CI/CD pipelines

Automated testing workflows

SASS/CSS

Pangealand S.L. / exoticca.com

Senior Software Engineer

2018 - 2019 · Barcelona, Spain

Development and architectural refactor of B2B front-end platform and supporting APIs, preparing client and server systems for global expansion and multi-locale support. Front-end integrated with RESTful services built on Node.js, backed by MongoDB and MySQL data stores.

Development and maintenance of internal operational dashboard used across multiple departments, with a focus on workflow optimization, platform usability, and scalable front-end/back-end architecture. Hands-on AWS and Linux server administration across the stack, including S3 for storage.

JavaScript

Angular

Node.js

MongoDB

MySQL

RESTful API integration

PHP

AWS (S3, hosting infrastructure)

Linux server administration

SASS/CSS

Econocom Spain / Digital Dimension

Full-stack Developer

2016 - 2017 · Barcelona, Spain

Front-end and back-end development of SaaS platform serving France and Spain. Integration of multiple external APIs and transformation of structured data for dashboard presentation and operational workflows.

Early large-scale React and Redux experience within agile Scrum environment, contributing to front-end architecture, responsive interface development, UX improvements, and modernization of application workflows.

React

Redux

Node.js

JavaScript

Docker

RESTful APIs

SASS/CSS

70 Main Street Studios

Front-end Developer / UX Designer

2015 - 2016 · Toronto, Canada

Front-end development, UX design, and responsive interface implementation for international clients including Scotiabank, TD Bank, Rogers, and Nickelodeon.

TECHNICAL PROJECTS

AI & Experimentation

LLAAB

508 commits

Learning Loop & Agent Automation Base. A local-first AI knowledge system for turning unstructured media into reviewed, source-backed knowledge artifacts. TypeScript pnpm/Turborepo monorepo with Vite + React Router SPA, Hono + Bun API server, Zod schemas, typed Hono RPC client/server contracts, and a file-based markdown vault of typed nodes. Supports YouTube transcript ingestion, LLM extraction, canonical-idea consolidation, durable RunNode traces, configurable LLM routing across OpenCode, Ollama, Anthropic, and LM Studio, plus MCP/CLI/operator surfaces. Recent work adds review-gated wiki generation: selected canonical ideas compile into vault drafts, then explicitly promote into knowledge/wikis/ with citations, revision-safe updates, derived graph exports, and no automatic Git mutation.

@finograc/ai-harness

v0.1.0 · 13 commits

Deterministic, composable pipeline for preprocessing inputs before LLM usage. Extract, reduce, structure, then optionally pass to an LLM. Designed to maximize the value of each token by doing as much work as possible before the model sees the input.

Full-stack Application

touch-monorepo

2,932 commits

Full-stack TypeScript monorepo for an IoT-connected product management and operational control system. React + React Router client with PandaCSS, Hono API server with Drizzle ORM + SQLite, hardware relay integration via USB HID, i18n with dynamic language support (EN/ES/CA), role-based auth, real-time timer and session management, and deployment tooling for Raspberry Pi. Uses pnpm workspaces, Turborepo, and consumes multiple @finograc packages in production.

monorepo-demo

101 commits

Live full-stack demo at https://finograc.github.io/monorepo-demo, built from monorepo-starter, showcasing a reusable TypeScript monorepo architecture with React, Hono, Auth.js, Drizzle ORM, i18n, and design-system components. API layer uses Hono RPC for end-to-end type-safe client/server contracts, with the typed client inferred directly from server route definitions. Server currently deployed to AWS App Runner; database migration from SQLite to Postgres in progress. Includes 3 interactive demos: AI-generated Markdown pipeline, Queensland TMR data visualisation, and supply-chain security scanner demo.

CLI & Developer Tools

@finograc/gli

v1.25.4 · 213 commits

Git CLI with live-updating terminal PR dashboard (like http for pull requests). Interactive rebase, branch selection, multi-repo config, clickable PR links. Built on the GitHub CLI.

Development of reusable front-end components, marketing platforms, and responsive web experiences within multidisciplinary agency environment, collaborating closely with design, development, and project teams.

- JavaScript
- Angular
- Ruby
- MySQL
- SASS/CSS
- Jenkins CI

Apple UK & Spain

Software Developer / Technical Support

2010 - 2012 · Barcelona, Spain

Promoted internally from technical support and sales into IT-focused role following independent development of an internal call-tracking and operational reporting application.

The platform improved workflow efficiency, operational visibility, and reporting for management and support teams. Later adopted for use by 300+ employees during high-volume Black Friday operations.

- JavaScript
- PHP
- MySQL
- CSS

Metropolis Media Group S.L.

Full-stack Developer, Designer, Product Manager

2003 - 2014 · Barcelona, Spain

Long-term contractor for front-end and back-end web development, UX design, digital consulting, product management, and print/media design across multiple client projects and industries.

Development of websites, custom CMS platforms, operational business tools, publishing systems, and online business platforms, alongside branding, UX design, and layout design of an upscale monthly lifestyle magazine.

- JavaScript
- PHP
- WordPress
- MySQL
- SASS/CSS
- Linux server administration

Architecture & Configuration

@finograc/genx

v5.40.3 · 745 commits

Project generator and codemod toolkit. Scaffolds new packages, migrates conventions (OXC, AI agent configs, git hooks, vitest), audits existing repos for drift, and keeps structure consistent across the ecosystem. The central tool that ties everything together.

@finograc/oxc-config

v2.8.2 · 188 commits

Shared oxfmt and oxlint configuration for the Rust-powered OXC toolchain. Composable presets for formatting (TypeScript, React, CSS, JSON, markdown) and linting (with role-specific presets for client, server, CLI, and library packages).

@finograc/deps-policy

v0.26.9 · 167 commits

Central dependency version policy consumed by genx and other tooling. Includes its own CLI for auditing, updating, and snapshotting dependency state across the ecosystem.

@finografic/cli-kit

v1.4.0 · 48 commits

Composable CLI primitives shared across all @finografic CLI tools. Flag parsing, interactive prompts, TUI table rendering, file diff display, XDG config paths. The single hard dependency for every CLI in the ecosystem.

EDUCATION

Ontario College of Art and Design University (OCAD U)

Design, Advertising Art Direction

Toronto, Canada

Multidisciplinary program covering core design principles, graphic design, multimedia, and communication in advertising. Emphasis on critical thinking, conceptual development, and collaboration.

LANGUAGES

- English : Native
- Spanish : Fluent (spoken + written)
- Catalan : Conversational

@finografic/core

v0.15.0 · 145 commits

Shared TypeScript utilities, type-level primitives (casing transforms, object key utilities, enum helpers), type guards, and time/string formatting functions.

@finografic/md-lint

v0.16.6 · 94 commits

Structural markdown linter with two scoped rule sets: strict rules for human-facing docs, relaxed rules for AI agent instruction files (CLAUDE.md, AGENTS.md). Auto-classifies files by path.

@finografic/project-scripts

v1.5.4 · 68 commits

Monorepo utility scripts: build artifact cleanup, SQLite database setup and migration, cross-platform deployment packaging, and GitHub release automation.

Front-end & Design System

@finografic/design-system

v1.19.2 · 434 commits

Token-driven React design system built on PandaCSS and Ark UI headless primitives. 20+ components (accordion, dialog, data-table, toast, tabs, forms), OKLCH color palette, semantic tokens, responsive viewport system, and a companion icon management tool.

@finografic/lucide-manager

v0.12.11 · 93 commits

Dev tool for managing a Lucide icon registry. Browser-based picker UI for selecting from 1500+ icons, generating strongly-typed TypeScript registry files for the design system package.

@finografic/zustand-context-creator

v1.2.0 · 47 commits

Lightweight utility for creating type-safe Zustand stores with React Context integration. Auto-generated setters, persistence, DevTools support. Inspired by patterns from Matt Pocock's TypeScript work.